

**Time:** 5 minutes**Outcome:** Apply the natural-rate hypothesis and vertical long-run Phillips curve.**Difficulty:** ●●● Advanced

Long-Run Phillips Curve



THE CORE IDEA

The long-run Phillips curve (LRPC) is vertical at the natural unemployment rate because expected inflation eventually adjusts. A demand expansion can temporarily reduce unemployment, but rising inflation expectations shift the short-run Phillips curve (SRPC) upward. Changes in job matching or skill mismatches can change the natural rate and shift LRPC.



HOW TO RECOGNIZE IT

- A demand expansion initially moves along the short-run curve.
- As inflation expectations rise, the short-run curve shifts upward and unemployment returns to the natural rate.
- Improved job matching can lower the natural rate, shifting LRPC left.

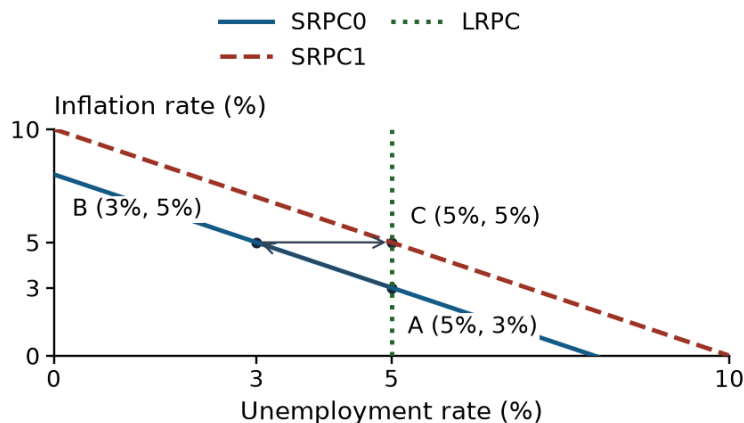


WATCH OUT

Do not treat a higher expected inflation rate as a permanent way to lower unemployment. Along the long-run Phillips curve, different inflation rates can coexist with the same natural rate of unemployment.



WORKED EXAMPLE: RETURNING TO THE NATURAL RATE



The economy starts at A: 5% unemployment and 3% inflation. A demand expansion moves it along SRPC0 to B, at 3% unemployment and 5% inflation. As expected inflation rises, SRPC shifts up to SRPC1. The final point C is on LRPC at 5% unemployment and 5% inflation. Unemployment returns to its natural rate; there is no permanent inflation-unemployment tradeoff.



CHECK YOURSELF

Why does higher inflation not permanently lower unemployment in the natural-rate model?

**READY?**

Return to the game and master this concept.